

Regression Math: - (Unit 8/2nd part)

Given below ages of 7 couple: -

Age of husbands	25	27	29	32	35	37	39
Age of wives	18	20	20	25	25	30	37

① Compute the regression equation age of wives on age of husbands.

Solution:

Let, x age of husbands and y denotes age of wives.

We know that,

$$① \quad c = a + by$$

$$② \quad Y = a_1 + b_1 x$$

$$③ \quad a_1 = \bar{y} - b_1 \bar{x}$$

$$④ \quad b_1 = \frac{\sum xy - \frac{\sum x \cdot \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$⑤ \quad x = a_2 + b_2 y$$

$$⑥ \quad a_2 = \bar{x} - b_2 \bar{y}$$

$$⑦ \quad b_2 = \frac{\sum xy - \frac{\sum x \cdot \sum y}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}}$$

Calculation of Regression coefficients:-

Age of Husb (X)	Age of Wives (Y)	xy	x ²	y ²
25	18	450	625	324
27	20	540	729	400
29	20	580	841	400
32	25	800	1024	625
35	25	875	1225	625
37	30	1110	1369	900
39	37	1443	1521	1369
$\Sigma x = 224$	$\Sigma y = 175$	$\Sigma xy = 5798$	$\Sigma x^2 = 7334$	$\Sigma y^2 = 4643$

$$\begin{aligned}
 \therefore \text{So, } b_1 &= \frac{\Sigma xy - \frac{\Sigma x \cdot \Sigma y}{n}}{\Sigma x^2 - \frac{(\Sigma x)^2}{n}} \\
 &= \frac{5798 - \frac{224 \times 175}{7}}{7334 - \frac{(224)^2}{7}} \\
 &= \frac{5798 - 5600}{7334 - 7168} \\
 &= 1.19
 \end{aligned}$$

$$\begin{aligned}
 \text{So, } a_1 &= \bar{y} - b_1 \bar{x} \\
 &= \frac{\Sigma y}{n} - b_1 \frac{\Sigma x}{n}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{175}{7} - 1.19 \left(\frac{224}{7} \right) \\
 &= 25 - (1.19 \times 32) \\
 &= -13.08.
 \end{aligned}$$

$$\text{So, } y = a_1 + b_1 x$$

$$= -13.08 + 1.19x \quad (\text{Ans:})$$

So, the regression equation ^{of age} of wives (y) on age of husbands (x) is expressed as follow:

$$-13 + 1.19x.$$

① Regression equation of age of husbands (x) on age of wives (y) is:—

$$\begin{aligned}
 \text{We have, } b_2 &= \frac{\sum xy - \frac{\sum x \cdot \sum y}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}} \\
 &= \frac{198}{4643 - \frac{(175)^2}{7}} \quad [\text{from ① no}] \\
 &= \frac{198}{4643 - 4375} = \frac{198}{268} = 0.74
 \end{aligned}$$

$$\begin{aligned}
 \text{So, } a_2 &= \bar{x} - b_2 \bar{y} \\
 &= \frac{\sum x}{n} - b_2 \frac{\sum y}{n}
 \end{aligned}$$

$$= \frac{224}{7} - 0.74 \left(\frac{175}{7} \right)$$

$$= 32 - 18.5$$

$$= 13.5$$

So, The regression equation of age of husbands (x) on age of wives (y) is :-

$$x = a_2 + b_2 y$$

$$= 13.5 + 0.74 y. \quad (\text{Ans})$$

iii) No! The most likely age of husbands for age of wife 35 years:-

$$X = a_2 + b_2 y$$

$$= 13.5 + (.74 \times 35)$$

$$= 39.4 \quad (\text{Ans})$$

iv) the most likely age of wife for age of husband 36 years.

$$y = a_1 + b_1 x$$

$$= -13.08 + (1.19 \times 36)$$

$$= -13.08 + 42.84$$

$$= 29.76 \quad (\text{Ans})$$

① Coefficient of correlation: —

$$\sqrt{b_1 \times b_2}$$

$$= \sqrt{1.19 \times 0.74}$$

$$= \sqrt{0.8806}$$

$$= 0.94 \text{ (Ans.)}$$

It expresses negative correlation.

①

2 NO. Math.

Age	50	63	45	36	72	65	47	25
Blood pressure	62	60	51	25	79	43	60	33

① Compute the regression equation of age on blood pressure.
Solution: on age.

We have,

$$c = a + by$$

$$y = a_1 + b_1 x$$

$$a_1 = \bar{y} - b_1 \bar{x}$$

$$b_1 = \frac{\sum xy - \frac{\sum x \cdot \sum y}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$x = a_2 + b_2 y$$

$$a_2 = \bar{x} - b_2 \bar{y}$$

$$b_2 = \frac{\sum xy - \frac{\sum x \cdot \sum y}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}}$$

Let,

x age of people and y denotes blood pressure.

So,

x	y	xy	x ²	y ²
50	62	3100	2500	3844
63	60	3780	3969	3600
45	51	2295	2025	2601
36	25	900	1296	625
72	79	5688	5184	6241
65	43	2795	4225	1849
47	60	2820	2209	3600
25	33	824	625	1089
$\Sigma x = 403$	$\Sigma y = 413$	$\Sigma xy = 22,202$	$\Sigma x^2 = 22,033$	$\Sigma y^2 = 23,449$

$$\begin{aligned}
 \therefore b_1 &= \frac{\Sigma xy - \frac{\Sigma x \cdot \Sigma y}{n}}{\Sigma x^2 - \frac{(\Sigma x)^2}{n}} \\
 &= \frac{22,202 - \frac{403 \times 413}{8}}{22,033 - \frac{(403)^2}{8}} \\
 &= \frac{22,202 - 20,804.88}{22,033 - 20,301.13} \\
 &= \frac{1,397.12}{1,731.87} \\
 &= 0.807
 \end{aligned}$$

$$\begin{aligned}
 a_1 &= \bar{y} - b_1 \bar{x} \\
 &= \frac{\sum y}{n} - b_1 \frac{\sum x}{n} \\
 &= \frac{403}{8} - 0.807 \times \frac{413}{8} \\
 &= 50.375 - 41.661 \\
 &= 8.714
 \end{aligned}$$

$$\begin{aligned}
 a_1 &= \bar{y} - b_1 \bar{x} \\
 &= \frac{\sum y}{n} - b_1 \frac{\sum x}{n} \\
 &= \frac{413}{8} - .807 \times \frac{403}{8} \\
 &= 51.625 - 40.653 \\
 &= 10.972
 \end{aligned}$$

∴ So, the regression equation of blood pressure on age is.

$$y = a_1 + b_1 x$$

$$\begin{aligned}
 &= 8.714 + 0.807x \\
 &= 10.972 + 0.807x \quad (\text{Ans})
 \end{aligned}$$

② We have,

$$\begin{aligned}
 b_2 &= \frac{\sum xy - \frac{\sum x \cdot \sum y}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}} \\
 &= \frac{1397.12}{23449 - \frac{(413)^2}{8}} \\
 &= \frac{1397.12}{23449 - 21221.125} = \frac{1397.12}{2127.88} \\
 &= 0.657.
 \end{aligned}$$

$$\begin{aligned}
 a_2 &= \bar{x} - b_2 \bar{y} \\
 &= \frac{\sum x}{n} - b_2 \frac{\sum y}{n} \\
 &= \frac{403}{8} - 0.657 \times \frac{413}{8} \\
 &= 50.375 - 33.918 \\
 &= 16.457,
 \end{aligned}$$

So, the regression equation of age on blood pressure is,

$$\begin{aligned}
 x &= a_2 + b_2 y \\
 &= 16.457 + 0.657 y \quad \text{Ans}
 \end{aligned}$$

(iii) The expected blood pressure of a person who is 52 years is:-

$$\begin{aligned}
 y &= a_1 + b_1 x \\
 &= 10.972 + 0.807 \times 52 \\
 &= 10.972 + 41.964 \\
 &= 52.94.
 \end{aligned}$$